# Chapter 7

# Unsteady Aerodynamic Flows

This chapter will examine the general low-speed unsteady potential flow problem. Specifically, we will revisit flow-field modeling in the unsteady case, and also the unsteady Bernoulli equation for the unsteady pressure. The special case of unsteady airfoil flows will be examined in more detail.

## 7.1 Unsteady Flow-Field Representation

Chapter 2 defined the source density and vorticity as the divergence and curl of the velocity field.

$$\begin{aligned} \sigma &= \nabla \cdot \mathbf{V} \\ \boldsymbol{\omega} &= \nabla \times \mathbf{V} \end{aligned}$$

These involve only spatial derivatives and hence apply instantaneously even if $\mathbf{V}$ is changing in time. Similarly, representation of the velocity field via its source, vorticity, and boundary contributions

$$\mathbf{V}(\mathbf{r},t) \;=\; \mathbf{V}_\sigma \;+\; \mathbf{V}_\omega \;+\; \mathbf{V}_b$$

involves only spatial integrations, and likewise applies instantaneously. An unsteady flow can therefore be represented in the same manner as a steady flow, but all the relevant quantities will now depend on time as well as space.

$$\mathbf{V}_\sigma(\mathbf{r},t) \;\equiv\; \frac{1}{4\pi} \iiint \sigma(\mathbf{r}',t)\, \frac{(\mathbf{r}-\mathbf{r}')}{|\,\mathbf{r}-\mathbf{r}'|^3}\, \mathrm{d}x'\, \mathrm{d}y'\, \mathrm{d}z' \tag{7.1}$$

$$\mathbf{V}_\omega(\mathbf{r},t) \;\equiv\; \frac{1}{4\pi} \iiint \boldsymbol{\omega}(\mathbf{r}',t) \times \frac{(\mathbf{r}-\mathbf{r}')}{|\,\mathbf{r}-\mathbf{r}'|^3}\, \mathrm{d}x'\, \mathrm{d}y'\, \mathrm{d}z' \tag{7.2}$$

$$\mathbf{V}_b \;=\; \mathbf{V}_\infty \qquad \text{(observer moving steadily in airmass)} \tag{7.3}$$

$$\mathbf{V}_b \;=\; \mathbf{0} \qquad \text{(observer fixed in airmass)} \tag{7.4}$$

We see that the unsteadiness of the velocity field is captured entirely by the time dependence of the source and vorticity fields $\sigma(\mathbf{r},t), \boldsymbol{\omega}(\mathbf{r},t)$. Furthermore, since the lumping process is strictly spatial, as in the steady case the integrals above can be simplified using the lumped unsteady sheet strengths $\lambda(s,\ell,t), \boldsymbol{\gamma}(s,\ell,t)$, line strengths $\Lambda(\ell,t), \boldsymbol{\Gamma}(\ell,t)$, or point strengths $\Sigma(t)$. For *free vorticity* such as in trailing wakes, an alternative approach is to move the vortex points rather than change the singularity strengths. In that case (7.2) is still valid, but now the integration points $\mathbf{r}'$ are functions of time.

In typical steady flow applications so far we have assumed that the body is fixed, so that the last boundary-condition component $\mathbf{V}_b$ is the freestream velocity $\mathbf{V}_\infty$. In unsteady flow applications it is often more convenient to represent the body motion explicitly via its velocity $\mathbf{U}(t)$ and rotation rate $\boldsymbol{\Omega}(t)$. In this case we choose $\mathbf{V}_b = \mathbf{0}$, and $\mathbf{V}$ then represents the "perturbation velocity," which is what's seen by an observer stationary with respect to the airmass.

## 7.2 Unsteady Potential Flow

The Helmholtz vorticity transport equation (1.95) dictates that aerodynamic flows have zero vorticity everywhere except in thin viscous layers adjacent to a solid body and in its trailing wakes. This conclusion remains valid for unsteady flows.

$$\boldsymbol{\omega} \equiv \nabla \times \mathbf{V} = \mathbf{0} \qquad \text{(outside viscous layers)} \tag{7.5}$$

Furthermore, the low-speed continuity equation is unchanged in the unsteady case,

$$\sigma \equiv \nabla \cdot \mathbf{V} = 0 \tag{7.6}$$

so that the velocity field still has zero divergence everywhere in the flow interior. The overall conclusion is that the physical constraints on the vorticity and source distributions are the same in both steady and unsteady low speed flows. The instantaneous velocity field outside the viscous layers (or the entire Equivalent Inviscid Flow, introduced in Chapter 3) can still be represented by the perturbation potential $\varphi(\mathbf{r},t)$.

$$\mathbf{V}(\mathbf{r},t) = \nabla\varphi \tag{7.7}$$

If source sheets and doublet sheets are used as in most panel methods, then the unsteady potential can be expressed explicitly via the usual superposition integrals, which now have unsteady sheet strength distributions.

$$\varphi(\mathbf{r},t) = \varphi_\lambda + \varphi_\mu \tag{7.8}$$

$$\varphi_\lambda(\mathbf{r},t) = \frac{1}{4\pi} \iint \lambda(s,\ell,t) \, \frac{-1}{|\mathbf{r}-\mathbf{r}'|} \, \mathrm{d}s \, \mathrm{d}\ell \tag{7.9}$$

$$\varphi_\mu(\mathbf{r},t) = \frac{1}{4\pi} \iint \mu(s,\ell,t) \, \frac{\hat{\mathbf{n}} \cdot (\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3} \, \mathrm{d}s \, \mathrm{d}\ell \tag{7.10}$$

## 7.3 Governing Equations for Unsteady Potential Flow

Figure 7.1 shows a rigid body moving through the fluid, seen in two different frames.

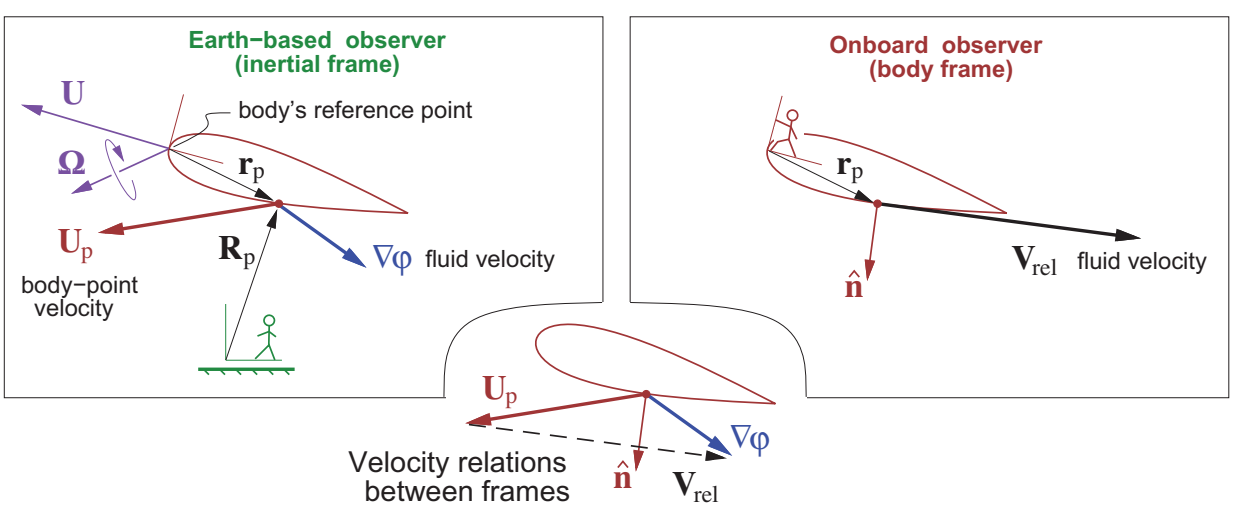


Figure 7.1: Positions and velocities seen by an Earth-based observer, and by an onboard observer.

The body's motion is defined by the velocity $\mathbf{U}(t)$ of some chosen reference point on the body, and by the body's rotation rate $\mathbf{\Omega}(t)$. An arbitrary point P on the body has location $\mathbf{r}_\mathrm{p}$ relative to the reference point, and $\mathbf{R}_\mathrm{p}$ relative to the ground observer. This point's velocity relative to the ground observer is then

$$\mathbf{U}_\mathrm{p} \;=\; \frac{\mathrm{d}\mathbf{R}_\mathrm{p}}{\mathrm{d}t} \;=\; \mathbf{U} \;+\; \mathbf{\Omega}\times\mathbf{r}_\mathrm{p} \tag{7.11}$$

which is the same as equation (6.3) considered earlier. The fluid velocity as seen by the ground observer is $\nabla\varphi$ as defined previously. The local fluid velocity seen in the local body frame at point $\mathbf{r}$ is then obtained by subtracting the body frame's local velocity.

$$\mathbf{V}_\mathrm{rel}(\mathbf{r},t) \;=\; \nabla\varphi \;-\; (\mathbf{U} + \mathbf{\Omega}\times\mathbf{r}) \tag{7.12}$$

The field equation and boundary conditions for the perturbation potential $\varphi(\mathbf{r},t)$ are

$$\boxed{\begin{aligned} \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2} \;\equiv\; \nabla^2\varphi \;&=\; 0 \\ \text{Solid-body BC:}\quad \mathbf{V}_\mathrm{rel}\cdot\hat{\mathbf{n}} \;&=\; 0 \\ \text{or equivalently:}\quad \nabla\varphi\cdot\hat{\mathbf{n}} \;&=\; (\mathbf{U} + \mathbf{\Omega}\times\mathbf{r})\cdot\hat{\mathbf{n}} \\ \text{Freestream BC:}\quad \varphi \;&\to\; 0 \end{aligned}} \tag{7.13}$$

which must all instantaneously hold for each instant in time. The time dependence arises through the body motion $\mathbf{U}(t)$ and $\mathbf{\Omega}(t)$, and possibly also through atmospheric motion and body deformation as will be described later.

### 7.3.1 Pressure calculation

Chapter 1 gave the derivation if the unsteady Bernoulli equation (1.105) for incompressible flow:

$$\frac{\partial\varphi}{\partial t} \;+\; \frac{1}{2}V^2 \;+\; \frac{p}{\rho} \;+\; gz \;=\; C \tag{7.14}$$

Here we have $V^2 = |\nabla\varphi|^2$, and we will also absorb the hydrostatic term $gz$ into $p/\rho$, as described in Section 1.9.3. The integration constant $C$ can be evaluated at any point where the pressure and velocity are known, such as at infinity where $p = p_\infty$, and where the fluid is at rest so that all $\varphi$ derivatives vanish.

$$p \;+\; \frac{1}{2}\rho\,|\nabla\varphi|^2 \;+\; \rho\,\frac{\partial\varphi}{\partial t} \;=\; p_\infty \tag{7.15}$$

It's essential to note that (7.15) as written applies only in an inertial frame of reference, since it was ultimately derived from the momentum equation (1.86) for which we set $\mathbf{f}$ to be gravity, omitting any non-inertial forces. This is why we chose $V^2 = |\nabla\varphi|^2$, and <u>not</u> $V^2 = |\mathbf{V}_\mathrm{rel}|^2$. Furthermore, the time derivative $\partial\varphi/\partial t$ must be performed at a spatial point which is inertial, meaning that this point is either stationary ($V_\infty = 0$) or at most translating with uniform speed ($V_\infty \neq 0$) relative to Earth. Specifically, the time derivative must <u>not</u> be performed at a point fixed on an accelerating or rotating body.

A practical complication here is that because the solution to the potential-flow problem as formulated above is given in the body coordinates in the form $\varphi(\mathbf{r},t)$, its explicit time derivative is at fixed $\mathbf{r}$ which is not an inertial point. However, we require a time derivative at fixed $\mathbf{R}$ which is inertial. The distinction between the time derivatives in the two frames is shown in Figure 7.2.

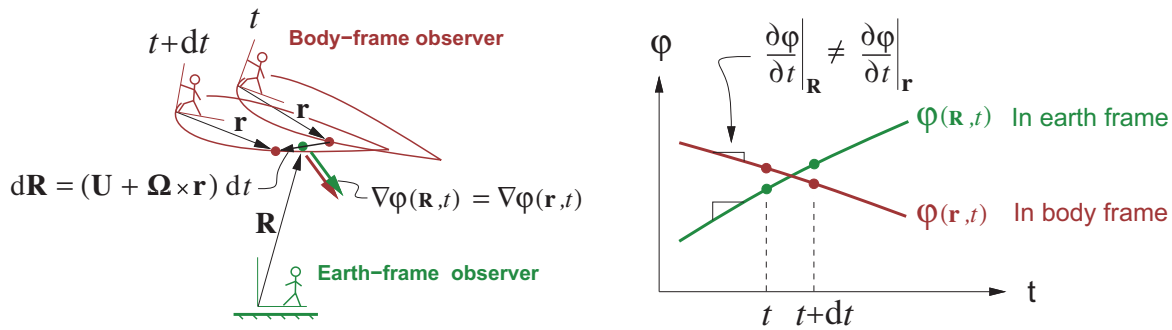


Figure 7.2: Potential-field time variation at a point $\mathbf{R}$ fixed in the Earth frame, and another point $\mathbf{r}$ fixed in the body frame. At the moment when the points coincide, the $\varphi$ and $\nabla\varphi$ values in the two frames are equal, but the time derivatives $\partial\varphi/\partial t$ are not.

To perform the frame conversion, we first note from Figure 7.1 that positions in the two frames are related by the correspondence function $\mathbf{R}_{(\mathbf{r},t)}$ which obeys

$$\left.\frac{\partial \mathbf{R}}{\partial t}\right|_{\mathbf{r}} = \mathbf{U} + \mathbf{\Omega}\times\mathbf{r}$$

which can then be used to relate time derivatives in the two frames. Considering the potential to be in the form $\varphi(\mathbf{R}(\mathbf{r},t),t)$, we obtain its time rate in the body frame via the chain rule,

$$\dot{\varphi} \equiv \left.\frac{\partial \varphi}{\partial t}\right|_{\mathbf{r}} = \left.\frac{\partial \varphi}{\partial t}\right|_{\mathbf{R}} + \left.\frac{\partial \varphi}{\partial \mathbf{R}}\right|_{t}\cdot\left.\frac{\partial \mathbf{R}}{\partial t}\right|_{\mathbf{r}} = \left.\frac{\partial \varphi}{\partial t}\right|_{\mathbf{R}} + \nabla\varphi\cdot(\mathbf{U}+\mathbf{\Omega}\times\mathbf{r}) \tag{7.16}$$

where $\partial\varphi/\partial\mathbf{R}$ is just a more intuitive way to write the spatial gradient $\nabla\varphi$ in this context. Relation (7.16) then gives the potential's time-rate in the Earth frame entirely in terms of quantities in the body frame.

$$\left.\frac{\partial \varphi}{\partial t}\right|_{\mathbf{R}} = \dot{\varphi} - \nabla\varphi\cdot(\mathbf{U}+\mathbf{\Omega}\times\mathbf{r}) \tag{7.17}$$

The pressure can now be expressed using the unsteady Bernoulli equation (7.15) applied in the frame of the Earth-based observer in Figure 7.2:

$$p_{(\mathbf{r},t)} = p_\infty - \tfrac{1}{2}\rho\,|\nabla\varphi|^2 - \rho\left.\frac{\partial \varphi}{\partial t}\right|_{\mathbf{R}} \tag{7.18}$$

$$\boxed{p_{(\mathbf{r},t)} = p_\infty - \tfrac{1}{2}\rho\,|\nabla\varphi|^2 + \rho\,\nabla\varphi\cdot(\mathbf{U}+\mathbf{\Omega}\times\mathbf{r}) - \rho\,\dot{\varphi}} \tag{7.19}$$

An equivalent alternative form is in terms of the body-relative velocity $\mathbf{V}_{\rm rel}$ given by (7.12).

$$p_{(\mathbf{r},t)} = p_\infty + \tfrac{1}{2}\rho\,|\mathbf{U}+\mathbf{\Omega}\times\mathbf{r}|^2 - \tfrac{1}{2}\rho\,|\mathbf{V}_{\rm rel}|^2 - \rho\,\dot{\varphi} \tag{7.20}$$

It's useful to note that in a steady flow situation we have

$$\begin{aligned}\mathbf{U} &= -\mathbf{V}_\infty\\ \mathbf{\Omega} &= \mathbf{0}\\ \dot{\varphi} &= 0\end{aligned}$$

in which case (7.20) reduces to the familiar steady Bernoulli equation.

$$p_{(\mathbf{r},t)} = p_\infty + \tfrac{1}{2}\rho V_\infty^2 - \tfrac{1}{2}\rho\,|\mathbf{V}_{\rm rel}|^2$$

## 7.4 Potential Jump of Unsteady Vortex Sheet

The physical zero pressure jump requirement across a vortex sheet was combined with the steady Bernoulli equation in Section 5.4 to obtain the convection condition (5.15) for the potential jump. For the unsteady case this needs to be re-examined because of the additional term in the unsteady Bernoulli equation (7.19). We therefore consider an unsteady vortex sheet shown in Figure 7.3 which has separate potentials and velocities on its upper and lower sides. The vortex sheet can represent either a thin lifting surface or a wake.

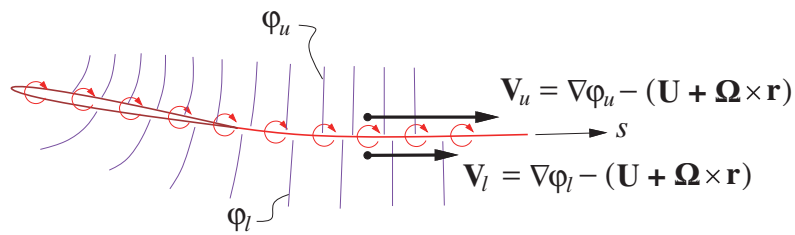


Figure 7.3: Perturbation potential isolines and airfoil-frame velocities above and below an unsteady vortex sheet, on the airfoil and on the wake.

As in the steady case, a thin unsteady wake has the same physical requirement of a zero static pressure jump $\Delta p \equiv p_l - p_u$ across it. Taking the difference of the unsteady pressure expression (7.20) between a lower and upper point across the sheet, and setting the result to zero gives

$$\begin{aligned} \Delta p = 0 \qquad \rightarrow \qquad \tfrac{1}{2}\left(\mathbf{V}_u \cdot \mathbf{V}_u \;-\; \mathbf{V}_l \cdot \mathbf{V}_l\right) \;+\; \left(\dot{\varphi}_u - \dot{\varphi}_l\right) &= 0 \\ \tfrac{1}{2}\left(\mathbf{V}_u + \mathbf{V}_l\right) \cdot \left(\mathbf{V}_u - \mathbf{V}_l\right) \;+\; \left(\dot{\varphi}_u - \dot{\varphi}_l\right) &= 0 \\ \mathbf{V}_a \cdot \widetilde{\nabla}(\Delta\varphi) \;+\; \dot{(\Delta\varphi)} &= 0 \end{aligned} \tag{7.21}$$

where $\mathbf{V}_u, \mathbf{V}_l$ are the upper and lower $\mathbf{V}_{\text{rel}}$ velocities, $\mathbf{V}_a$ is the average *sheet velocity*

$$\mathbf{V}_a = \tfrac{1}{2}\left(\mathbf{V}_u + \mathbf{V}_l\right)$$

and $\widetilde{\nabla}()$ is the surface gradient within the sheet. Note that (7.21) reduces to the earlier steady-flow potential jump result (5.15) if we set $\dot{(\Delta\varphi)} = 0$.

### 7.4.1 Potential-jump convection

The unsteady wake zero pressure jump condition (7.21) can be rewritten as a substantial derivative.

$$\begin{aligned} \frac{\partial(\Delta\varphi)}{\partial t} \;+\; \mathbf{V}_a \cdot \widetilde{\nabla}(\Delta\varphi) &= 0 \\ \text{or} \qquad \frac{\mathrm{D}(\Delta\varphi)}{\mathrm{D}t} &= 0 \qquad \text{(on wake)} \end{aligned} \tag{7.22}$$

Equation (7.22) implies that the wake potential jump $\Delta\varphi(s,\ell,t) \equiv \varphi_u - \varphi_l$ , set initially at the trailing edge by the Kutta condition, convects unchanged at the sheet velocity $\mathbf{V}_a$, as sketched in Figure 7.4.

### 7.4.2 Shed vorticity

From Figure 7.3 we see that the wake sheet has a tangential velocity jump $\mathbf{V}_u - \mathbf{V}_l = \widetilde{\nabla}(\Delta\varphi)$, which corresponds to a vortex sheet of the following strength.

$$\gamma \;=\; \hat{\mathbf{n}} \times \left(\mathbf{V}_u - \mathbf{V}_l\right) \;=\; \hat{\mathbf{n}} \times \widetilde{\nabla}(\Delta\varphi) \tag{7.23}$$

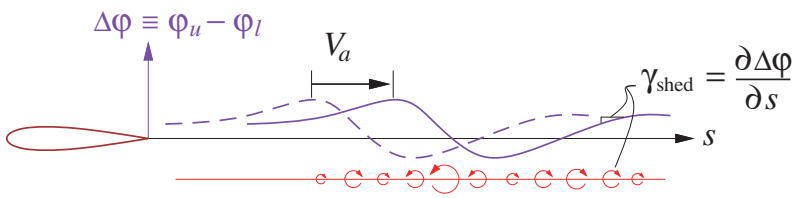


Figure 7.4: Potential jump $\Delta\varphi$ across a wake convects without change at the average sheet velocity. If the flow is unsteady, the resulting streamwise gradients in $\Delta\varphi$ imply the presence of shed vorticity perpendicular to $\mathbf{V}_a$.

The *trailing vorticity* is the streamwise component of this $\boldsymbol{\gamma}$ vector, and the *shed vorticity* is the transverse component. Choosing the $s$ coordinate to be parallel to $\mathbf{V}_a$,

$$\hat{\mathbf{s}} \equiv \frac{\mathbf{V}_a}{|\mathbf{V}_a|} \tag{7.24}$$

gives the following explicit definitions of the trailing and shed vorticity components.

$$\gamma_{\text{trail}} \equiv \hat{\mathbf{s}} \cdot \boldsymbol{\gamma} \tag{7.25}$$

$$\gamma_{\text{shed}} \equiv (\hat{\mathbf{s}} \times \boldsymbol{\gamma}) \cdot \hat{\mathbf{n}} = \hat{\mathbf{s}} \cdot \widetilde{\nabla}(\Delta\varphi) = \frac{\partial \Delta\varphi}{\partial s} \tag{7.26}$$

Comparing equation (7.26) with (7.21) together with the $\hat{\mathbf{s}}$ definition shows that $\gamma_{\text{shed}}$ can be alternatively given by the time rate of $\Delta\varphi$.

$$\gamma_{\text{shed}} = -\frac{(\dot{\Delta\varphi})}{|\mathbf{V}_a|} \begin{cases} = 0 \quad , & \text{steady flow} \\ \neq 0 \quad , & \text{unsteady flow} \end{cases} \tag{7.27}$$

Shed vorticity is therefore present only in unsteady flows, and is a major complication in the formulation of unsteady potential flow calculation methods.

## 7.5 Unsteady Flow Categories

As derived above, unsteady potential flow introduces three basic new effects not present in steady flow: unsteady $\nabla\varphi \cdot \hat{\mathbf{n}}$ due to body motion $\mathbf{U}(t), \boldsymbol{\Omega}(t)$, the additional $\rho\,\partial\varphi/\partial t$ term in the Bernoulli equation for the pressure, and the generation of shed vorticity $\gamma_{\text{shed}}$. Only some of these effects may be important in any given situation, giving a natural categorization of unsteady flows summarized in Table 7.1.

Table 7.1: Unsteady flow regimes and retained or discarded effects

| | $\partial(\nabla\varphi \cdot \hat{\mathbf{n}})/\partial t$ | $\rho\,\partial\varphi/\partial t$ | $\gamma_{\text{shed}}$ |
|---|---|---|---|
| Steady | 0 | 0 | 0 |
| Quasi-steady I | $\surd$ | neglected | neglected |
| Quasi-steady II | $\surd$ | neglected | estimated |
| Unsteady | $\surd$ | $\surd$ | $\surd$ |

The quasi-steady approximation is employed for defining the aerodynamic loads in flight in Chapter 6, where the unsteady forces on a maneuvering aircraft are assumed to be the same as those in steady flight at the same instantaneous flight condition defined by the parameters $\alpha, \beta, \bar{p}, \bar{q}, \bar{r}$. In the "Quasi-steady II" case a slight refinement is made by estimating the shed vorticity effects via the additional $\dot{\bar{\alpha}}$ parameter.

## 7.6 Unsteady Panel Method

Standard panel methods which solve the steady potential flow problem can be extended to the unsteady case. The source+doublet panel method will be used here to illustrate these necessary modifications.

### 7.6.1 Sources of unsteadiness

Figure 7.5 shows the possible origins of flow unsteadiness in an aerodynamic flow which must be captured by any general unsteady flow solution method. These are further described below.

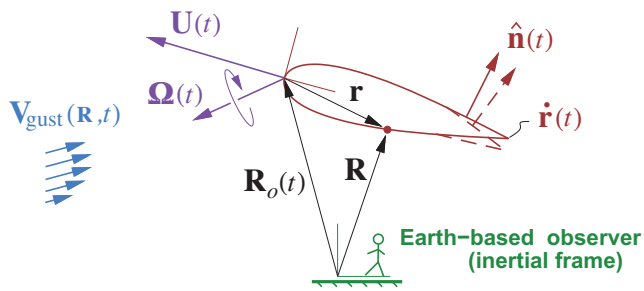


Figure 7.5: Typical sources of unsteadiness in an aerodynamic flow are unsteady body motion $\mathbf{U}(t), \mathbf{\Omega}(t)$, unsteady body deformation $\hat{\mathbf{n}}(t), \dot{\mathbf{r}}(t)$, and a spatially-varying or unsteady atmospheric velocity field $\mathbf{V}_{\text{gust}}(\mathbf{R},t)$.

#### Body motion

One possible source of unsteadiness is unsteady rigid-body motion. This is defined by the specified velocity and rotation rate $\mathbf{U}(t)$ and $\mathbf{\Omega}(t)$ introduced in the body-point velocity definition (7.11).

#### Gust field

Unsteady flow can also be caused by a nonuniform atmospheric motion or "gust" velocity field $\mathbf{V}_{\text{gust}}(\mathbf{R},t)$ which is specified. The instantaneous Earth position of any body point $\mathbf{r}$ on the aircraft is

$$\mathbf{R}(\mathbf{r},t) = \mathbf{R}_{\text{o}}(t) + \mathbf{r}$$

where $\mathbf{R}_{\text{o}}$ is obtained by integrating $\mathbf{U}$ in time, as described in Chapter 9. The gust velocity at the body point $\mathbf{r}$ is then

$$\mathbf{V}_{\text{gust}}(\mathbf{r},t) = \mathbf{V}_{\text{gust}}(\mathbf{R}(\mathbf{r},t),t)$$

where Earth/body axis conversions using the body Euler angles would be used, as described in Section 9.4. These are obtained by integration of the aircraft rotation rate $\mathbf{\Omega}(t)$. In reality, any such gust field will be modified by the presence of the body, but for the typical gust field which has long variations relative to the body size this effect will be small and is generally neglected. This is consistent with classical thin airfoil approximations.

#### Body deformation

The third cause of flow unsteadiness is a time-dependent body geometry with local velocity $\dot{\mathbf{r}}$, such as an oscillating flap or a flapping wing. If the deformation of the motion is small, its effects can be approximated by holding the geometry fixed, but still using the actual surface velocity $\dot{\mathbf{r}}$ and normal vectors $\hat{\mathbf{n}}$ when forming the flow-tangency condition.

### 7.6.2 Wake convection

A major complication in calculation of unsteady flows is the evolution of the wake, both its doublet strength as well as its geometry. The physically correct wake shape is the streakline (in 2D) or streaksurface (in 3D) attached to the trailing edge. An example is shown in Figure 7.6 for a heaving airfoil. In the most physically accurate free wake case the wake shape's evolution in time depends on the history of the velocity field of the airfoil+wake combination, and hence depends on the flow solution itself. This renders the problem nonlinear, since the kernel functions for the wake doublet sheet, in particular the $\mathbf{r}'$ values in the $\varphi_\mu$ integral (7.10), will now depend on the solution.

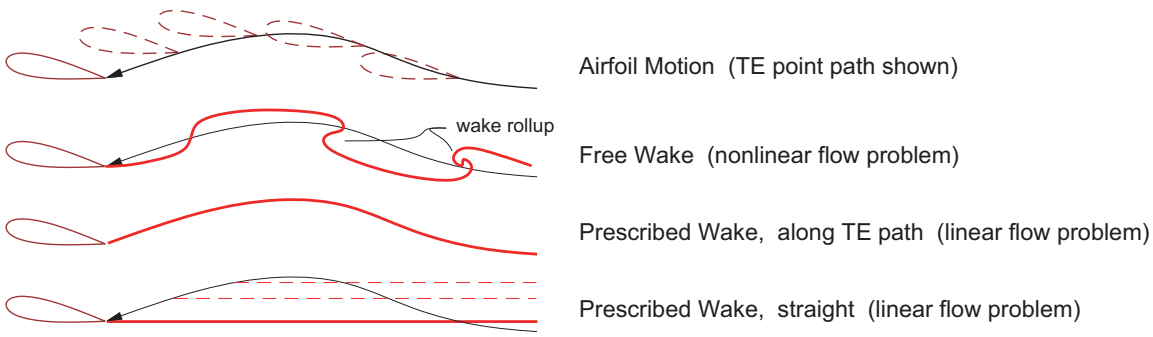


Figure 7.6: Three different wake geometry models. In free wake model, spanwise wake roll-up typically occurs at locations of strong shed vorticity.

A further complication is that a free vortex sheet will in general undergo *roll-up* wherever the vorticity is concentrated. Steady flow problems which only have trailing vorticity typically have wake roll-up at the wake sheet's wingtip edges, as sketched in Figure 5.4. Unsteady flow problems in addition also have roll-up caused by shed vorticity. The extreme deformation of the wake shape undergoing roll-up makes accurate tracking of this shape difficult and computationally expensive.

These complications associated with a free wake can be mostly eliminated by using the *prescribed-wake assumption*. Here the convection of the wake by the velocity field is ignored, and instead the wake geometry is prescribed explicitly. Two possible choices are shown in Figure 7.6. The simplest straight-wake assumption is quite adequate for modest airfoil motions, and is almost invariably used by Vortex-Lattice methods. The great advantage of a prescribed wake is that it renders the potential flow problem linear, since the kernel functions then become independent of the flow solution. Unless the rolled-up wake closely interacts with a downstream surface, the neglect of roll-up generally results in very minor errors in a potential flow solution.

### 7.6.3 Panel method formulation

Figure 7.7 shows the velocity field relative to the body, both on the exterior and the interior. These are defined as

$$\mathbf{V}(\mathbf{r},t) = -\mathbf{U} - \mathbf{\Omega}\times\mathbf{r} - \dot{\mathbf{r}} + \mathbf{V}_{\text{gust}} + \nabla\varphi \qquad \text{(physical exterior velocity)} \tag{7.28}$$

$$\mathbf{V}_{\text{int}}(\mathbf{r},t) = -\mathbf{U} - \mathbf{\Omega}\times\mathbf{r} - \dot{\mathbf{r}} + \mathbf{V}_{\text{gust}} \qquad \text{(nonphysical interior velocity)} \tag{7.29}$$

where only $\nabla\varphi$ is unknown a priori. The physical flow-tangency requirement on the exterior is

$$\mathbf{V}\cdot\hat{\mathbf{n}} = 0 \tag{7.30}$$

so that the source sheet strength is

$$\lambda \equiv (\mathbf{V} - \mathbf{V}_{\text{int}})\cdot\hat{\mathbf{n}} = -\mathbf{V}_{\text{int}}\cdot\hat{\mathbf{n}} \tag{7.31}$$

which is known and thus can be explicitly specified for each point on the body surface.

If we now specify a zero interior potential, $\varphi_{\text{int}} = 0$, the doublet sheet strength then becomes the physical external perturbation potential $\varphi$, while on the wake it represents the physical jump $\Delta\varphi$.

$$\mu \equiv \varphi - \varphi_{\text{int}} = \varphi \tag{7.32}$$

$$\mu_{\text{w}} = \Delta\varphi \tag{7.33}$$

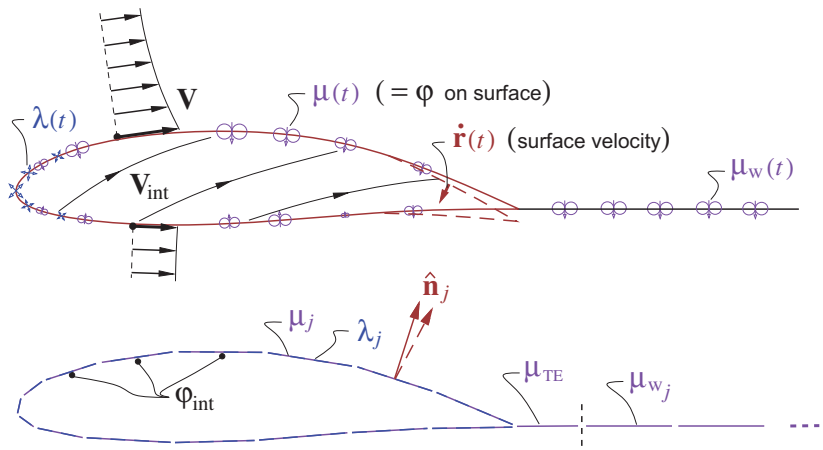


Figure 7.7: Body-relative exterior and interior velocities represented by body motion, gust field, and surface source and doublet sheets, shown in top figure. Panel discretization of the sheet strengths is shown in the bottom figure.

The zero interior potential is imposed via Green's identity

$$\varphi_{\text{int}} = -\frac{1}{2}\mu + \frac{1}{4\pi}\iint \mu \nabla\left(\frac{1}{r}\right)\cdot\hat{\mathbf{n}}\ \mathrm{d}\mathcal{S} + \frac{1}{4\pi}\iint \lambda \frac{-1}{r}\ \mathrm{d}\mathcal{S} + \frac{1}{4\pi}\iint \mu_{\text{w}} \nabla\left(\frac{1}{r}\right)\cdot\hat{\mathbf{n}}\ \mathrm{d}\mathcal{S}_{\text{w}} = 0 \tag{7.34}$$

which after evaluation on the paneled geometry becomes the following matrix equation for the unknown surface-panel doublet strengths $\mu_j$.

$$A_{ij}\,\mu_j = -A^{\text{w}}_{ij}\,\mu_{\text{w}_j} - B_{ij}\lambda_j \tag{7.35}$$

The trailing edge wake panel strength at each spanwise location is an unknown, but is related to the two top and bottom surface strengths at the trailing edge at that same location.

$$\mu_{\text{w}_{TE}} = (\mu_{j_{TE}})_u + (\mu_{j_{TE}})_l \tag{7.36}$$

This is equivalent to a Kutta condition.

The unsteady solution proceeds in a sequence of time steps, with the following operations performed at each time step:

1. Set the source panel strengths $\lambda_j$ from the known $\mathbf{V}_{\text{int}(\mathbf{r},t)}$.
2. Set all the wake-panel doublet strengths $\mu_{\text{w}_j}$ from the previous time step, by convecting $\mu_{\text{w}}$ without change along $\mathbf{V}_a$, according to the zero wake-load condition.

   $$\frac{\mathrm{D}\mu_{\text{w}}}{\mathrm{D}t} = 0$$

   A common approximation made here is to assume

   $$\mathbf{V}_a \simeq -(\mathbf{U} + \mathbf{\Omega}\times\mathbf{r})$$

   which is computationally simpler and much less expensive to evaluate.

3. Evaluate the righthand side in (7.35) using the known $\lambda_j$ and $\mu_{w_j}$
4. Solve linear system (7.35) for $\mu_j$, using the same LU-factored $A_{ij}$ matrix for all time steps.
5. Using $\mu_j, \mu_{w_j}, \lambda_j$, evaluate $\dot{\varphi}$, $\nabla\varphi$, $\mathbf{V}$ on surface, compute pressure and airloads.

For a much more detailed description of the necessary calculations see Katz and Plotkin [4].

The 3D unsteady panel method described above is quite general, but provides little insight into unsteady flow behavior except via numerical experimentation. Hence we will next consider simplified models of several specific unsteady flows of interest. Suitable approximations will be used as needed to obtain more concise semi-analytical results for unsteady aerodynamic forces and moments.

## 7.7 Unsteady 2D Airfoil

The unsteady 2D airfoil problem is important in many applications, such as prediction of gust loads on wings, cyclic loads on rotor blades, and the onset of flutter. See Bisplinghoff et al [53] for an overview. Here we will consider the case of a rigid airfoil moving at a uniform horizontal speed $U$, while undergoing unsteady heaving $h(t)$ and pitching $\theta(t)$ displacements about a baseline position and angle of attack $\alpha_0$, as shown in Figure 7.8. Note that the heave displacement $h$ is defined positive down.

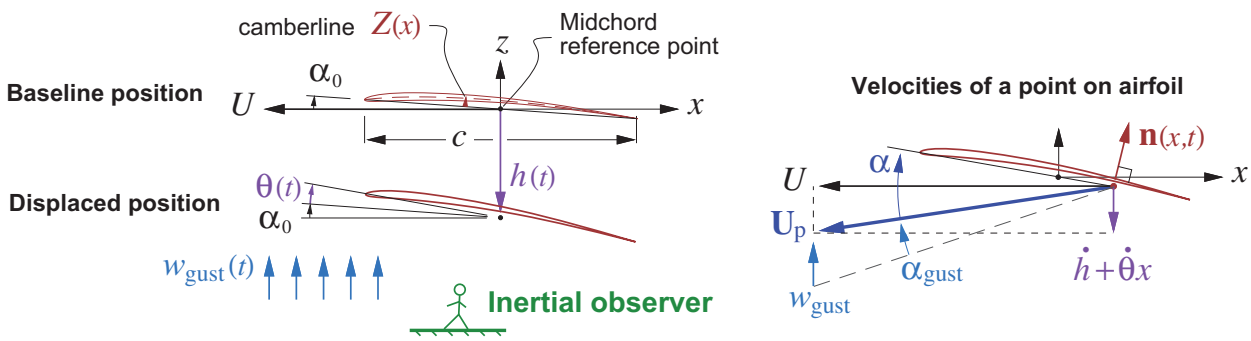


Figure 7.8: Airfoil moving at uniform horizontal baseline velocity $U$. Heave and pitch displacements $h(t), \theta(t)$ from baseline position give net resultant vertical velocity $\dot{h} + \dot{\theta}x$ at a some point $x$, which then define the local angle of attack $\alpha(x,t)$. Some vertical gust velocity $w_{\text{gust}}(t)$ produces an additional angle of attack $\alpha_{\text{gust}}$.

### 7.7.1 Geometric relations

The assumptions of first-order thin airfoil theory will be employed here. The airfoil camberline shape is $Z(x)$, with a negligibly small thickness. The geometric and flow angles are assumed to be small.

$$Z' \ll 1 \quad , \qquad \alpha_0 \, , \, \theta \ll 1 \quad , \qquad \dot{h} \, , \, \dot{\theta}c \, , \, w_{\text{gust}} \ll U$$

Following historical convention for unsteady airfoil analysis, the airfoil extends over $-c/2 \le x \le c/2$, so the reference point $x = 0$ is at midchord, as shown in Figure 7.8.

### 7.7.2 Problem formulation

The problem will be formulated in the frame and the axes of the inertial Earth-based observer. The camberline's normal vector therefore rotates with the airfoil.

$$\mathbf{n}(x,t) \;=\; \left(\alpha_0 + \theta - Z'\right)\hat{\mathbf{x}} + \hat{\mathbf{z}} \tag{7.37}$$

The small-angle assumptions give the following total velocity of a point on the airfoil.

$$\mathbf{U}_{\mathrm{p}(x,t)} \;=\; -U\hat{\mathbf{x}} - (\dot{h} + \dot{\theta}x)\hat{\mathbf{z}} \tag{7.38}$$

The perturbation velocity field is represented by a vortex sheet of strength $\gamma(x)$ on the airfoil and wake. The wake portion is absent in steady thin airfoil theory (Appendix D), but is needed here to represent shed vorticity. In the Earth frame, the total vortex sheet strength gives the following instantaneous vertical fluid velocity $w$ at any given location on the $x$-axis, which adds to any $w_{\mathrm{gust}}$ already present.

$$w_{(x,t)} \;=\; \frac{1}{2\pi}\int_{-c/2}^{\infty} -\gamma_{(x',t)}\,\frac{\mathrm{d}x'}{x-x'} \tag{7.39}$$

In the airfoil frame, the corresponding fluid velocity at location $x$ is then

$$\mathbf{V}_{\mathrm{rel}} \;=\; (w + w_{\mathrm{gust}})\,\hat{\mathbf{z}} \;-\; \mathbf{U}_{\mathrm{p}} \;=\; U\hat{\mathbf{x}} \;+\; \Big(\dot{h} + \dot{\theta}x + w + w_{\mathrm{gust}}\Big)\,\hat{\mathbf{z}} \tag{7.40}$$

which is then used in the usual flow tangency condition on the camberline.

$$\mathbf{V}_{\mathrm{rel}}\cdot\mathbf{n} \;=\; 0 \qquad\qquad (-c/2 \le x \le c/2\,) \tag{7.41}$$

This applies only on the airfoil. In the wake, we must instead impose the zero pressure jump condition, by taking the $x$-derivative of the $\Delta\varphi$ convection equation (7.22), which then becomes the $\gamma$ convection equation. The governing equations over the entire vortex sheet $-c/2 \le x < \infty$ are then stated as follows.

$$\frac{1}{2\pi}\int_{-c/2}^{\infty}\gamma_{(x',t)}\,\frac{\mathrm{d}x'}{x-x'} \;=\; U\alpha_0 \;-\; UZ' \;+\; \dot{h} \;+\; (U\theta + \dot{\theta}x) \;+\; w_{\mathrm{gust}} \qquad (-c/2 \le x \le c/2\,) \tag{7.42}$$

$$\frac{\partial\gamma}{\partial t} \;+\; U\,\frac{\partial\gamma}{\partial x} \;=\; 0 \qquad\qquad (\;c/2 \le x < \infty\;) \tag{7.43}$$

The pressure loading required to calculate the lift and moment is obtained from equation (7.21). This in turn requires the time rate of the potential jump, which is defined in terms of the vortex sheet strength time rate.

$$\Delta p_{(x,t)} \;=\; \rho\,(\dot{\Delta\varphi}) \;+\; \rho U\gamma \tag{7.44}$$

$$(\dot{\Delta\varphi})_{(x,t)} \;=\; \int_{-c/2}^{x}\dot{\gamma}_{(x',t)}\,\mathrm{d}x' \tag{7.45}$$

Because equations (7.42) and (7.43) are both linear in $\gamma$, their solution $\gamma_{(x,t)}$ and corresponding $\Delta\varphi_{(x,t)}$ and $\Delta p_{(x,t)}$ can be decomposed into five independent components, with each component determined entirely by one of the five righthand side source terms in (7.42).

$$\gamma_{(x,t)} \;=\; \gamma_\alpha(x) \;+\; \gamma_Z(x) \;+\; \gamma_h(x,t) \;+\; \gamma_\theta(x,t) \;+\; \gamma_{\mathrm{gust}}(x,t) \tag{7.46}$$

The first two contributions $\gamma_\alpha, \gamma_Z$ are steady since they depend only on the fixed baseline angle of attack $\alpha_0$ and camberline shape $Z(x)$, respectively. These are treated by standard steady first-order thin airfoil theory, as derived in Appendix D. This result for the steady case is

$$\left.\begin{aligned} c_\ell \;&=\; c_{\ell_0} \;+\; 2\pi\alpha_0 \\ c_{m_{c/4}} \;&=\; c_{m_0} \end{aligned}\right\} \qquad \text{(steady baseline)}$$

where $c_{\ell_0}$ and $c_{m_0}$ depend only on the camberline shape. In the subsequent sections we will ignore these steady contributions and focus only on the three remaining unsteady parts driven by $h_{(t)}, \theta_{(t)}, w_{\mathrm{gust}(t)}$ which determine $\gamma_h, \gamma_\theta, \gamma_{\mathrm{gust}}$, respectively, and also their corresponding unsteady lift and moment.

### 7.7.3 Canonical impulse solutions

A solution to equations (7.42) and (7.43) have been obtained by Wagner [54] based on a unit step in $\dot{h}, \theta, \dot{\theta}$ corresponding to a sudden change of airfoil motion in heave rate, pitch, and pitch rate. Küssner [55] obtained a solution for the case of a unit step of $w_{\text{gust}}$, corresponding to the airfoil flying into a uniform vertical gust field with a sharp boundary.

In the Wagner case an important quantity is the angle of attack relative to the camber line at the $\frac{3}{4}$-chord location, or $x = c/4$.

$$\alpha_{3c/4} = \frac{\dot{h}}{U} + \theta + \frac{\dot{\theta} c}{4U} \tag{7.47}$$

In the Küssner case the relevant quantity is the apparent angle of attack resulting from the vertical gust velocity shown in Figure 7.8, which is assumed to act everywhere on the chord.

$$\alpha_{\text{gust}} = \frac{w_{\text{gust}}}{U} \tag{7.48}$$

If these quantities have step jumps of $\Delta\alpha_{3c/4}$ and/or $\Delta\alpha_{\text{gust}}$ at $t=0$, then the lift is

$$\frac{L'}{\frac{1}{2}\rho U^2 c} \equiv c_\ell(\bar{t}) = c_{\ell_Q} + c_{\ell_A} \tag{7.49}$$

$$c_{\ell_Q}(\bar{t}) = 2\pi\,\Delta\alpha_{3c/4}\,\Phi(\bar{t}) + 2\pi\,\Delta\alpha_{\text{gust}}\,\Psi(\bar{t}) \tag{7.50}$$

$$c_{\ell_A}(\bar{t}) = 2\pi\left[\frac{1}{2}\frac{\Delta\dot{h}}{U} + \frac{1}{2}\Delta\theta\right]\delta(\bar{t}) \tag{7.51}$$

$$\bar{t} \equiv \frac{U\,t}{c/2} \tag{7.52}$$

where $\Phi(\bar{t})$ is the *Wagner function*, $\Psi(\bar{t})$ is the *Küssner function*, and $\delta(\bar{t})$ is the unit impulse function. Their argument $\bar{t}$ is a non-dimensionalized time, which can also be interpreted as the distance that the airfoil has moved, in units of half-chord.

The lift is seen to be the sum of two parts:

1) $c_{\ell_Q}$ is the "quasi-steady" or "circulatory" part associated with vorticity shedding and circulation, and evolves in time according to the Wagner and Küssner functions shown in Figure 7.9.

2) $c_{\ell_A}$ is an "apparent-mass" or "impulsive" part associated with the instantaneous acceleration of the fluid immediately adjacent to the airfoil. A step change in velocity or angle produces an infinite acceleration, resulting in the impulsive lift.

Both the Wagner and Küssner functions asymptote to unity in the limit $t \to \infty$. Therefore, the general solution (7.50) asymptotes to the value $c_{\ell_\infty} = 2\pi(\Delta\alpha_{3c/4} + \Delta\alpha_{\text{gust}})$, as expected from steady thin airfoil theory.

The exact Wagner and Küssner functions cannot be expressed in terms of elementary functions, but are available in tabulated form. For calculations, the following curve-fit expressions are fairly accurate and convenient.

$$\Phi(\bar{t}) = 1 - 0.165\exp(-0.045\bar{t}) - 0.335\exp(-0.3\bar{t}) \tag{7.53}$$

$$\Psi(\bar{t}) = 1 - 0.5\exp(-0.13\bar{t}) - 0.5\exp(-1.0\bar{t}) \tag{7.54}$$

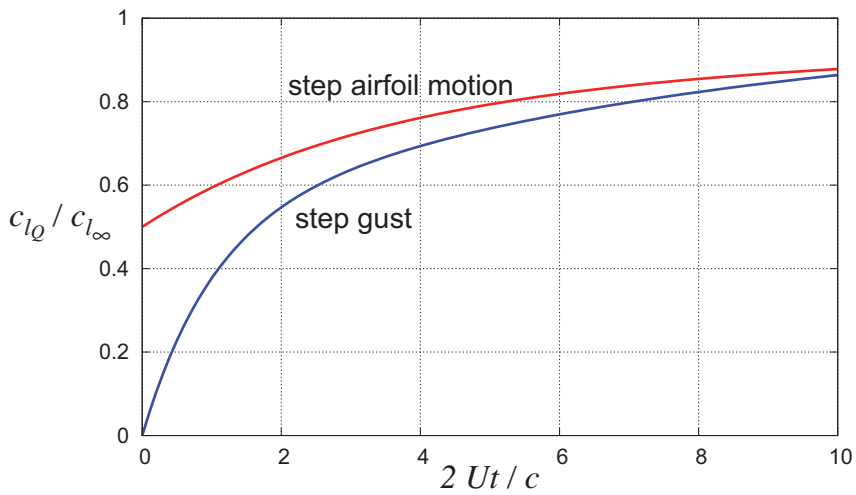


Figure 7.9: Circulatory part of lift coefficient versus time for a step increase in $\alpha_{3c/4}$ due to airfoil motion (Wagner solution), and a step increase in $\alpha_{\text{gust}}$ due to a gust velocity (Küssner solution). Total lift for the airfoil motion case contains an additional impulsive component, not shown.

### 7.7.4 General motion solution

The airfoil lift for a general motion can be obtained by Duhamel's (convolution) integral, which is a superposition of the infinitesimal step solutions over the motion history.

$$\begin{aligned} c_{\ell_Q}(\bar{t}) &= 2\pi\,\alpha_{3c/4}(0)\,\Phi(\bar{t}) \;+\; 2\pi\int_{0+}^{\bar{t}} \frac{\mathrm{d}\alpha_{3c/4}}{\mathrm{d}\bar{t}'}\,\Phi(\bar{t}-\bar{t}')\,\mathrm{d}\bar{t}' \\ &+ 2\pi\,\alpha_{\text{gust}}(0)\,\Psi(\bar{t}) \;+\; 2\pi\int_{0+}^{\bar{t}} \frac{\mathrm{d}\alpha_{\text{gust}}}{\mathrm{d}\bar{t}'}\,\Psi(\bar{t}-\bar{t}')\,\mathrm{d}\bar{t}' \qquad (7.55) \\ c_{\ell_A}(\bar{t}) &= 2\pi\left[\frac{c}{4}\frac{\ddot{h}}{U^2} + \frac{c}{4}\frac{\dot{\theta}}{U}\right] \qquad (7.56) \end{aligned}$$

The inputs are the airfoil motion given by $h(t)$ and $\theta(t)$, and the atmospheric gust motion given by $w_{\text{gust}}(t)$. These quantities define $\alpha_{3c/4}(t)$ and $\alpha_{\text{gust}}(t)$ which are then used in the convolution integrals for $c_{\ell_Q}$ above, and are also used directly for $c_{\ell_A}$.

### 7.7.5 Apparent mass

It is useful to note that the dimensional lift/span corresponding to $c_{\ell_A}$ given by (7.56) is

$$\begin{aligned} L'_A &= \tfrac{1}{2}\rho U^2\, c\, c_{\ell_A} \;=\; m'_A\,(\ddot{h} + U\dot{\theta}) \qquad (7.57) \\ m'_A &\equiv \rho\,\pi(c/2)^2 \qquad (7.58) \end{aligned}$$

where $m'_A$ is the *apparent mass* (per unit span), which is the mass of a unit-span cylinder of fluid whose diameter is airfoil chord. Hence, subjecting an airfoil to a normal acceleration results in an additional reaction force as though this cylindrical mass of the air was attached to the airfoil and also accelerated. The effective normal acceleration, in addition to $\ddot{h}$, also includes a Coriolis-like term $U\dot{\theta}$ due to airfoil rotation.

### 7.7.6 Sinusoidal motion solution

Theodorsen [56] derived solutions for the case of small-magnitude sinusoidal motion at some frequency $\omega$, specifically for use in predicting wing flutter. This situation is most conveniently treated by the introduction

of complex variables,

$$h(t) \;=\; h_0\,\cos(\omega t+\varphi_h) \;=\; \mathrm{Re}\{\bar{h}\,\mathrm{e}^{\mathrm{i}\omega t}\} \qquad ; \qquad \bar{h} \;=\; h_0\,\mathrm{e}^{\mathrm{i}\varphi_h} \tag{7.59}$$

$$\theta(t) \;=\; \theta_0\,\cos(\omega t+\varphi_\theta) \;=\; \mathrm{Re}\{\bar{\theta}\,\mathrm{e}^{\mathrm{i}\omega t}\} \qquad ; \qquad \bar{\theta} \;=\; \theta_0\,\mathrm{e}^{\mathrm{i}\varphi_\theta} \tag{7.60}$$

where the convenient complex amplitudes $\bar{h}, \bar{\theta}$ incorporate both the heave and pitch amplitudes $h_0, \theta_0$, and also the heave and pitch phase angles $\varphi_h, \varphi_\theta$. The notation $\mathrm{Re}\{\ \}$ denotes the real part of a complex quantity.

Substitution of the sinusoidal $h(t)$ and $\theta(t)$ expressions above into the general-motion Wagner solutions (7.55) and (7.56), gives the following solution for the lift coefficient, again as a sum of quasi-steady and apparent-mass contributions. The pitching moment coefficient can also be obtained.

$$\frac{L'}{\frac{1}{2}\rho U^2 c} \;\equiv\; c_\ell \;=\; \mathrm{Re}\left\{C(k)\,c_{\ell_Q} \;+\; c_{\ell_A}\right\} \tag{7.61}$$

$$c_{\ell_Q} \;=\; 2\pi\left[\theta+\frac{\dot{h}}{U}+\frac{c}{4}\frac{\dot{\theta}}{U}\right] \;=\; 2\pi\left[\bar{\theta}+\mathrm{i}k\frac{2\bar{h}}{c}+\mathrm{i}k\frac{\bar{\theta}}{2}\right]\mathrm{e}^{\mathrm{i}\omega t} \;=\; 2\pi\,\alpha_{3c/4} \tag{7.62}$$

$$c_{\ell_A} \;=\; 2\pi\left[\frac{c}{4}\frac{\dot{\theta}}{U}+\frac{c}{4}\frac{\ddot{h}}{U^2}\right] \;=\; 2\pi\left[\mathrm{i}k\frac{\bar{\theta}}{2}-k^2\frac{\bar{h}}{c}\right]\mathrm{e}^{\mathrm{i}\omega t} \tag{7.63}$$

$$\frac{M'}{\frac{1}{2}\rho U^2 c^2} \;\equiv\; c_{m_{c/2}} \;=\; \mathrm{Re}\left\{C(k)\,c_{m_Q} \;+\; c_{m_A}\right\} \tag{7.64}$$

$$c_{m_Q} \;=\; \frac{1}{4}\,c_{\ell_Q} \tag{7.65}$$

$$c_{m_A} \;=\; 2\pi\left[-\frac{c}{16}\frac{\dot{\theta}}{U}-\frac{c^2}{128}\frac{\ddot{\theta}}{U^2}\right] \;=\; 2\pi\left[-\mathrm{i}k\frac{\bar{\theta}}{8}+k^2\frac{\bar{\theta}}{32}\right]\mathrm{e}^{\mathrm{i}\omega t} \tag{7.66}$$

$$k \;\equiv\; \frac{\omega c}{2U} \;=\; \pi\,\frac{\text{chord}}{\text{distance traveled in one period}} \tag{7.67}$$

The dimensionless parameter $k$ which appears in this Theodorsen solution is the *reduced frequency*, which was previously mentioned in Section 1.5.3. It affects the strength of the shed vorticity, and also is a measure of the relative importance of the $\dot{\varphi}$ term in the unsteady Bernoulli equation (7.19). The other relevant parameters which appear are the maximum heave/chord ratio $h_0/c$, the maximum pitch displacement $\theta_0$, and the relative phase $\varphi_h - \varphi_\theta$ between the heave and pitch motions.

For finite values of $k$, the circulatory parts of the lift and moment in (7.61) and (7.64) are modified by the *Theodorsen Lag Function* $C(k)$ factor. This is a complex quantity, and is explicitly defined in terms of standard and modified Bessel functions $J_0, J_1$ and $Y_0, Y_1$, respectively.

$$H_0(k) \;=\; J_0(k) \;-\; \mathrm{i}\,Y_0(k) \qquad , \qquad H_1(k) \;=\; J_1(k) \;-\; \mathrm{i}\,Y_1(k)$$

$$C(k) \;=\; \frac{H_1(k)}{H_1(k)+\mathrm{i}H_0(k)} \tag{7.68}$$

Its real and imaginary components and its magnitude and phase angle are plotted in Figure 7.10.

In the quasi-steady limit $k \to 0$ the lag function becomes unity, $C(0) = 1+0\mathrm{i}$, and also all the time-rate terms in the unsteady lift and moment expressions become negligible. The resulting lift and moment coefficients then reduce to $c_\ell = 2\pi\alpha$ , $c_{m_{c/2}} = c_\ell/4$, which matches thin airfoil theory.

For $k > 0$ we have $|C(k)| < 1$ and $\angle C(k) < 0$, so the lift contribution of $c_{\ell_Q}$ will now be reduced in magnitude and will also have a phase lag. The main origin of these effects is the convection of shed vorticity into the wake. This contributes a vertical velocity $w$ at the airfoil which tends to oppose the current angle of attack,

and thus reduces the lift magnitude. But this vertical velocity contribution is delayed by the vorticity's convection time into the wake, which results in the phase lag.

As in the general-motion result (7.56), the remaining $c_{\ell_A}$ and $c_{m_A}$ "apparent-mass" parts act instantaneously. Because of their extra time derivatives these introduce phase leads into the lift and moment, but become significant only at larger $k$ values. Figure 7.11 shows the relative magnitudes of these various contributions to the lift for sinusoidally-heaving airfoils at two different reduced frequencies.

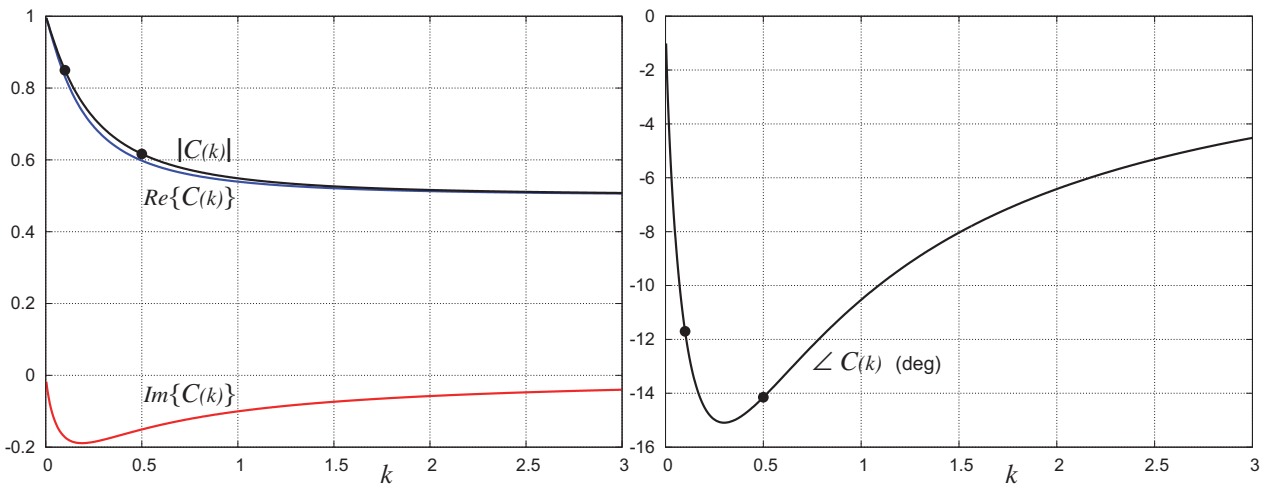


Figure 7.10: Real and imaginary parts, and magnitude and phase of the Theodorsen lag function. Magnitude and phase lag values at $k = 0.1$ and $k = 0.5$ are used in the example in Figure 7.11.

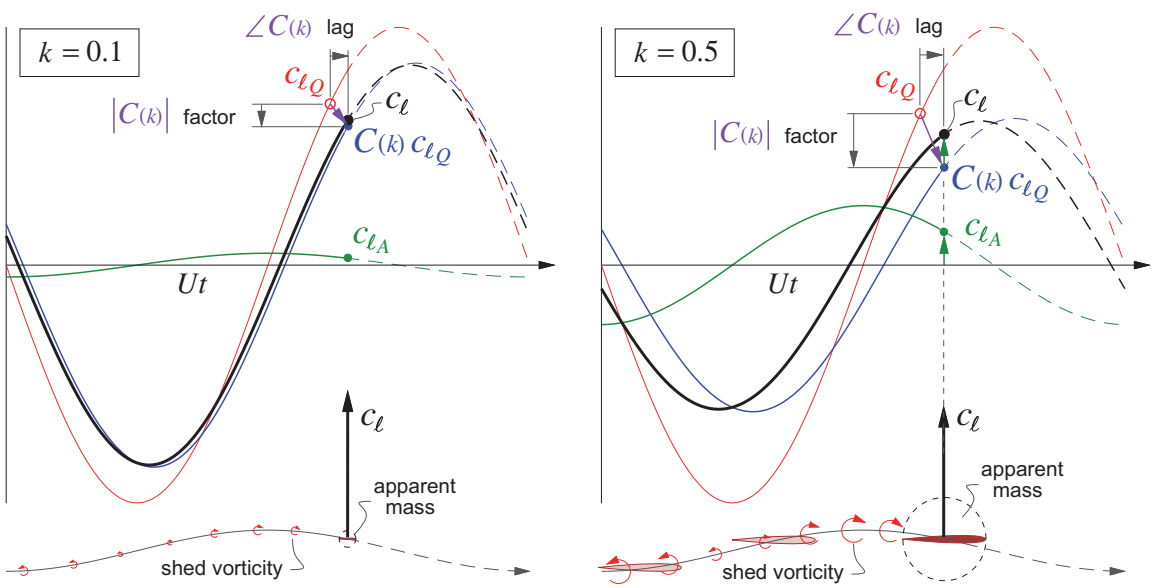


Figure 7.11: Unsteady lift coefficient components of airfoils undergoing sinusoidal heaving at zero pitch angle. Only the physical (real) parts are shown. The two airfoils have the same heave amplitude and wavelength, but different chords and hence different reduced frequencies $k = \pi \times$chord/wavelength. The quasi-steady lift $c_{\ell_Q} = 2\pi\alpha_{3c/4}$ is lagged and reduced in magnitude by the $C(k)$ Theodorsen function associated with shed vorticity. The impulsive lift contribution $c_{\ell_A}$ is the inertial reaction of the fluid's apparent mass subjected to vertical acceleration, and leads the quasi-steady lift by 90°.